EV RANGE PREDICTION

FROM VEHICLE SPECIFICATIONS

TECHNICAL REPORT

TECHTRACK 3.0 - EV-FOCUSED ML CASE BATTLE

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1. Abstract

This project develops a machine learning-based system for predicting the driving range of electric vehicles from static vehicle specifications. The dataset contains 478 EV models from 59 brands and includes battery, performance, charging, dimensional, drivetrain, body-type, seating, towing, and cargo-related specifications. The workflow includes data quality analysis, cleaning, exploratory data analysis, feature selection, reproducible preprocessing, model comparison, hyperparameter tuning, and held-out test evaluation. Nine regression approaches were investigated, including linear models, tree-based methods, Support Vector Regression (SVR), Gradient Boosting, and XGBoost. SVR was selected as the final model because it provided strong typical accuracy while controlling larger prediction errors. On the held-out test set, the selected SVR achieved an MAE of 10.309 km, RMSE of 13.646 km, and R^2 of 0.9824. The trained preprocessing and model pipeline was saved as a Joblib artifact and integrated into a Streamlit application for interactive range prediction.

Key Result

Final model: Support Vector Regression (SVR)
MAE: 10.309 km | RMSE: 13.646 km | R^2: 0.9824
Interactive deployment target: Streamlit

2. Introduction

Electric vehicles are becoming an important part of modern transportation, and driving range is one of the most visible specifications considered by users. A reliable range estimate can support vehicle selection, trip planning, charging decisions, and product comparison.

Driving range is influenced by several vehicle characteristics. Static specifications such as battery capacity, vehicle dimensions, power and torque-related characteristics, charging capability, drivetrain, body type, and seating configuration can provide useful predictive signals. Machine learning can learn these relationships from historical EV specifications and produce an estimated range for a new vehicle specification profile.

The objective of this project is therefore to build a reproducible regression system that predicts EV driving range in kilometres from static specifications while avoiding target leakage and providing an interactive testing interface.

3. Problem Statement and Scope

The task is to predict the target variable `range_km` from a set of static electric-vehicle specifications. The solution must include data cleaning, exploratory analysis, feature handling, regression model development, evaluation, reproducibility, and an interactive prediction mechanism.

Problem Definition

Input: 17 selected static EV specification features.
Output: Predicted driving range in kilometres (`range_km`).
Task type: Supervised regression.
Primary evaluation metrics: MAE, RMSE, and R^2.

The dataset does not contain real-time telemetry such as traffic, weather, HVAC usage, battery state of charge, battery health, road conditions, or driving behaviour. These factors are therefore outside the scope of the current model.

4. Dataset Description

The provided dataset contains 478 EV model records across 59 brands. Each row represents one EV model and includes 22 original columns covering performance, battery, charging, capacity, dimensions, drivetrain, body type, and metadata.

Property	Value
Number of EV records	478
Number of brands	59
Original columns	22

Target variable	range_km
Problem type	Regression

4.1 Key Dataset Fields

Feature	Type	Role
battery_capacity_kWh	Numerical	Input
top_speed_kmh	Numerical	Input
acceleration_0_100_s	Numerical	Input
drivetrain	Categorical	Input
car_body_type	Categorical	Input
range_km	Numerical	Target
efficiency_wh_per_km	Numerical	Excluded from final model

5. Data Cleaning and Quality Analysis

The initial audit identified missing values, mixed-type content in the cargo-volume column, high-cardinality metadata, and deliberate edge cases that required domain-aware handling. No exact duplicate rows were identified.

5.1 Missing Values

Feature	Missing count	Missing %
number_of_cells	202	42.26%
towing_capacity_kg	26	5.44%
torque_nm	7	1.46%
cargo_volume_l	4	0.84%
model	1	0.21%
fast_charge_port	1	0.21%
fast_charging_power_kw_dc	1	0.21%

Numerical missing values were handled using median imputation inside the model pipeline. Categorical missing values were handled using most-frequent imputation. This keeps preprocessing reproducible and prevents information from the test set from being used during training.

5.2 Invalid Cargo Values

The `cargo_volume_l` field contained non-numeric entries expressed as “Banana Boxes”. These values were treated as invalid numerical observations and converted to missing values rather than being manually guessed. Median imputation then handled the resulting missing entries.

5.3 Domain-Aware Outlier Review

Extreme-looking values were not automatically removed. Examples such as a large number of battery cells, high acceleration time, and zero towing capacity were reviewed as potentially legitimate EV specifications. This avoided deleting valid domain observations solely because they were numerically unusual.

5.4 Leakage Prevention

The feature `efficiency_wh_per_km` was excluded from the final prediction pipeline. It was retained for exploratory analysis and sanity checking only. This was done to prevent target leakage and to comply with the challenge requirement that this feature not be used as a final model input. The target `range_km` was also kept entirely separate from the predictor set.

6. Exploratory Data Analysis

Exploratory analysis focused on the target distribution, numerical relationships, feature-to-feature correlations, missingness, and categorical differences in observed range. The analysis was used to inform feature retention rather than simply to produce visualizations.

6.1 Numerical Summary

Feature	Mean	Median	Range
range_km	393.18	397.50	135 - 685 km
battery_capacity_kWh	74.04	76.15	21.3 - 118 kWh
top_speed_kmh	185.49	180	125 - 325 km/h
acceleration_0_100_s	6.88	6.60	2.2 - 19.1 s
fast_charging_power_kw_dc	125.01	113	29 - 281 kW
torque_nm	498.01	430	113 - 1350 Nm
length_mm	4678.51	4720	3620 - 5908 mm
width_mm	1887.36	1890	1610 - 2080 mm

6.2 Correlation with Range

The strongest numerical relationships with range were observed for battery capacity (+0.880), top speed (+0.732), fast charging power (+0.721), torque (+0.652), and acceleration time (-0.712). Height also showed a moderate negative relationship (-0.414), while width and length were moderately positive.

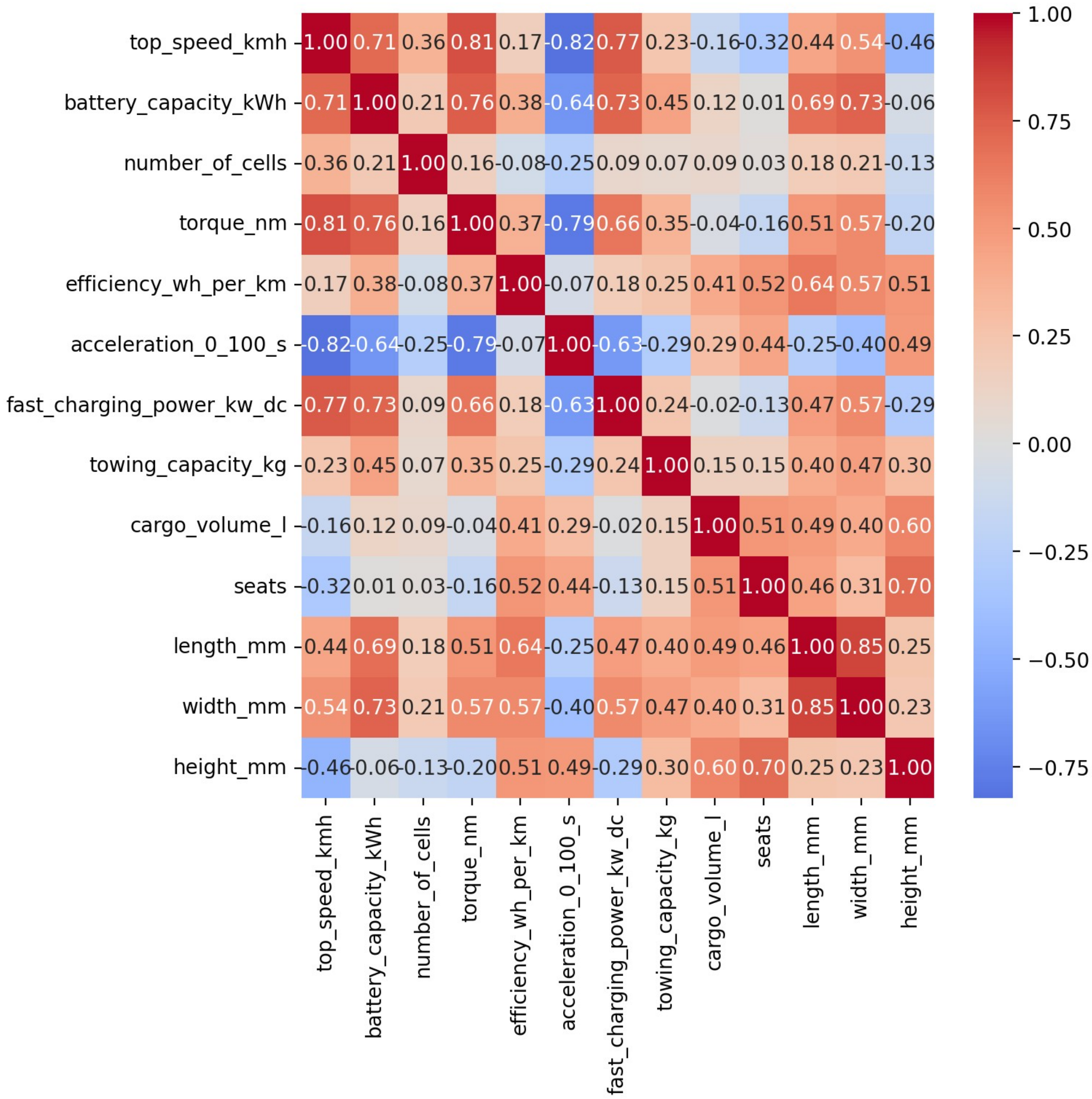

Figure 1. Correlation matrix of numerical features used during EDA.

Some predictors were strongly correlated with each other, including length and width (+0.85), top speed and acceleration time (-0.82), and top speed and torque (+0.81). These relationships were retained for model comparison because the project evaluates multiple model families and the final pipeline is empirically validated rather than based on correlation thresholds alone.

6.3 Categorical Findings

Drivetrain showed clear differences in observed range: AWD vehicles had a mean range of about 453.8 km, RWD about 418.2 km, and FWD about 298.0 km. Body type and segment also showed meaningful differences in average range, supporting their retention as categorical predictors.

7. Feature Selection

The final predictor set contains 17 features: 12 numerical and 5 categorical. Feature selection was based on data quality, cardinality, variance, metadata status, leakage prevention, and empirical validation.

Feature / group	Decision and rationale
model	Removed - nearly every model value is unique, creating very high cardinality.
battery_type	Removed - all 478 records are Lithium-ion, so the feature has zero variance.
source_url	Removed - metadata rather than a meaningful vehicle specification.
efficiency_wh_per_km	Removed from final model - target leakage prevention.
range_km	Target, not a predictor.
number_of_cells	Retained - ablation experiment showed measurable performance loss when removed.

7.1 Number-of-Cells Ablation Experiment

Configuration	MAE (km)	RMSE (km)	R^2
With number_of_cells	10.309	13.646	0.9824
Without number_of_cells	10.813	14.095	0.9812

Despite approximately 42.3% missingness, the feature contributed useful predictive signal after median imputation. Removing it increased MAE by about 0.50 km and RMSE by about 0.45 km, so it was retained.

8. Methodology and Reproducible Pipeline

An 80:20 train-test split was created with `random_state=42`. The training subset contained 382 rows and the held-out test set contained 96 rows. Five-fold cross-validation was used for model comparison and hyperparameter tuning. The test set was reserved for final evaluation.

End-to-End Workflow

Raw EV data -> audit -> cleaning -> EDA -> feature selection -> 80/20 split -> preprocessing -> cross-validation/tuning -> final test evaluation -> saved pipeline -> Streamlit demo

8.1 Preprocessing

- Numerical features: median imputation.
- Categorical features: most-frequent imputation followed by one-hot encoding.
- Unknown categorical levels: ignored through `OneHotEncoder(handle_unknown="ignore")`.
- SVR pipeline: preprocessing, sparse-compatible scaling, then SVR.
- All preprocessing is contained inside a single pipeline to keep cross-validation leakage-free.

9. Models Evaluated and Hyperparameter Tuning

Nine regression approaches were evaluated: Linear Regression, Ridge, Lasso, Elastic Net, Support Vector Regression (SVR), Decision Tree, Random Forest, Gradient Boosting, and XGBoost. Regularized linear models and nonlinear models were tuned using five-fold cross-validation. Grid Search or Randomized Search was selected according to the size of the search space.

Model	Best CV MAE (km)
SVR	12.004
XGBoost	12.492
Gradient Boosting	13.085
Linear Regression	14.646
Elastic Net	14.841
Lasso	14.873
Ridge	14.988
Random Forest	16.384
Decision Tree	19.520

9.1 Final SVR Hyperparameters

Parameter	Selected value
C	10000
epsilon	2
gamma	0.001

SVR was the strongest cross-validation model, while XGBoost and Gradient Boosting were close alternatives. The final decision was then checked on the previously untouched test set.

10. Final Model and Evaluation

The final held-out test evaluation compared SVR, Gradient Boosting, and XGBoost because these were the strongest candidates from cross-validation.

Model	MAE (km)	RMSE (km)	R^2
Gradient Boosting	9.966	13.822	0.9819
SVR	10.309	13.646	0.9824
XGBoost	10.353	14.481	0.9802

Final Model Selection

Selected model: SVR
Held-out MAE: 10.309 km
Held-out RMSE: 13.646 km
Held-out R^2: 0.9824
Maximum absolute error: 41.19 km

Gradient Boosting achieved the lowest MAE by a small margin of 0.343 km. SVR, however, achieved the lowest RMSE, the highest R^2, and a substantially smaller maximum absolute error. Because large prediction errors are undesirable for range estimation, SVR was selected as the more balanced final model.

11. Error and Residual Analysis

Residual analysis was performed using residual = actual range - predicted range. This provides information about the direction, magnitude, and pattern of model errors beyond aggregate metrics.

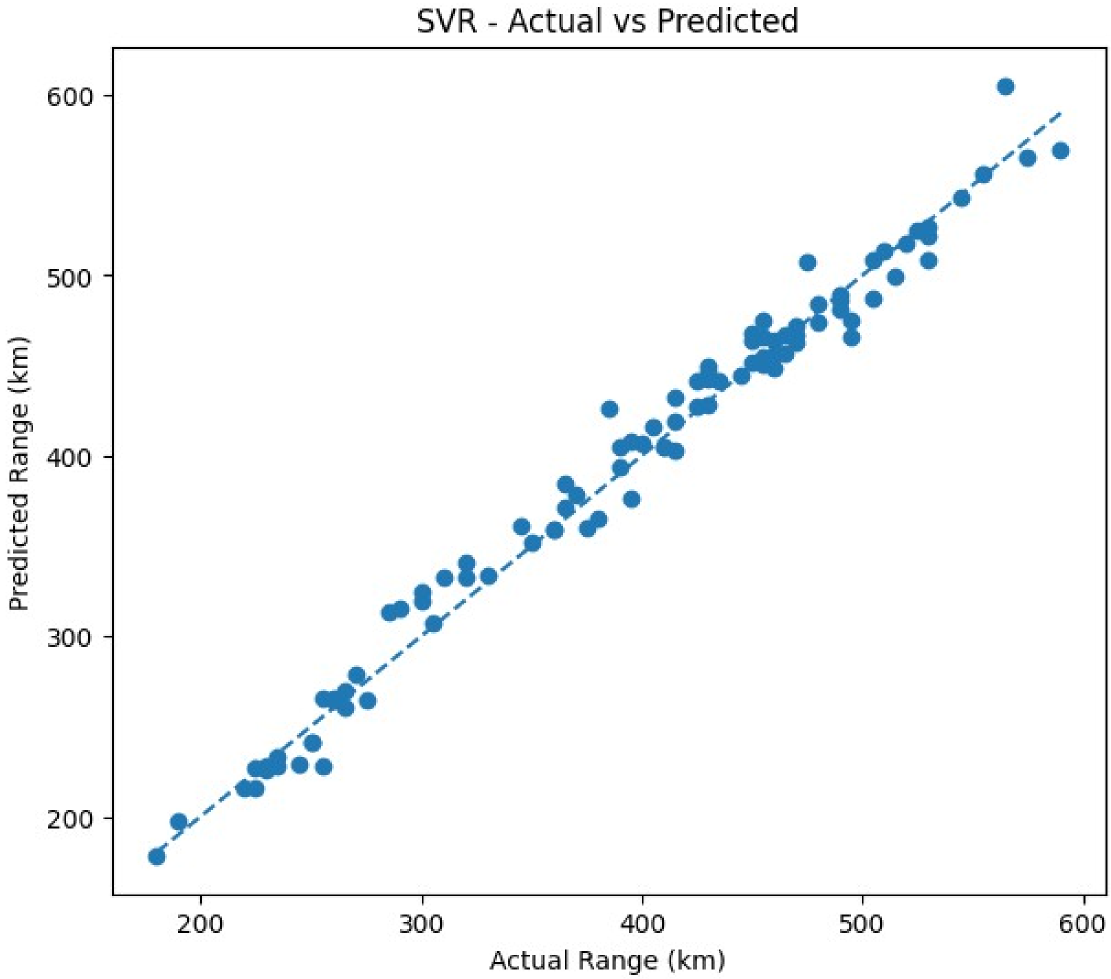

Figure 2. SVR actual versus predicted range on the held-out test set.

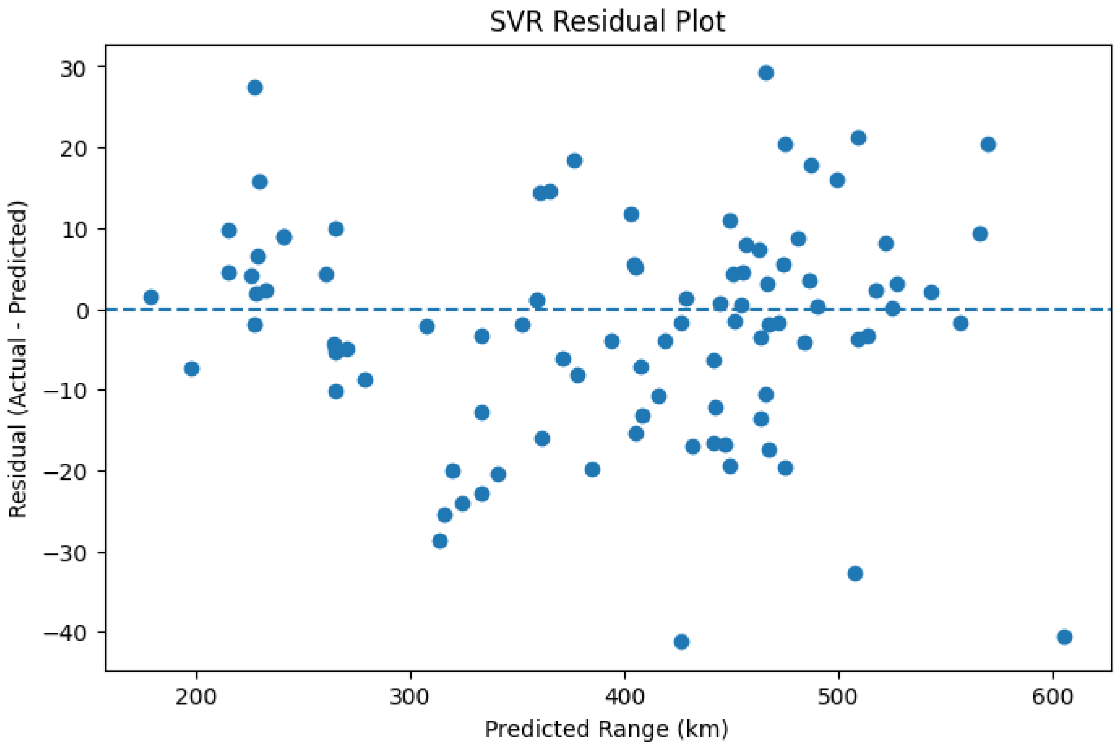

Figure 3. SVR residual plot.

The residuals are broadly scattered around the zero line without an obvious strong systematic curve. A slight negative mean residual (-2.23 km) indicates mild average overprediction. The largest absolute SVR error was 41.19 km. The residual spread appears somewhat wider at higher predicted ranges, suggesting greater uncertainty for some high-range EVs.

11.1 Direction of Prediction Errors

On the held-out test set, 47.92% of SVR predictions were underestimates and 52.08% were overestimates. This is an observed property of this test split rather than a competition requirement. The project uses MAE, RMSE, and R^2 as the primary quantitative evaluation criteria.

11.2 Worst-Case Error Review

Example EV	Actual range	Predicted range	Absolute error
Kia	385	426.19	41.19
Lucid	565	605.50	40.50
Porsche	475	507.64	32.64
Hyundai	495	465.76	29.24
Volkswagen	285	313.67	28.67

These cases were not removed as outliers because the underlying EV specifications are plausible. They represent difficult observations for the learned relationship rather than obvious data-entry errors.

12. Model Explainability

Because SVR does not expose tree-style feature importance, permutation importance was used. The analysis measures how much predictive performance changes when the values of a feature are randomly permuted on the test data.

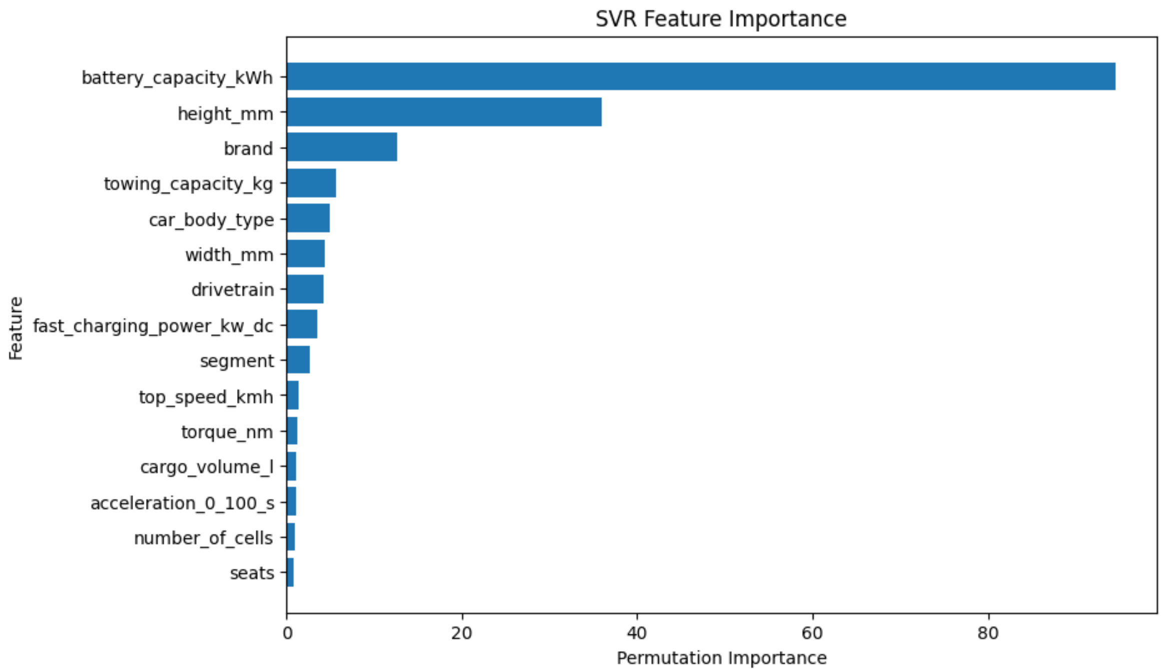

Figure 4. SVR permutation importance on the held-out test set.

Battery capacity was by far the dominant predictive feature in the permutation analysis (importance approximately 94.54), followed by height, brand, towing capacity, body type, width, drivetrain, and charging power. These scores indicate predictive contribution rather than causality; correlated vehicle specifications can share predictive information.

13. Interactive Streamlit Application

A Streamlit application was developed to provide live prediction from the saved model pipeline. The interface groups the 17 model inputs into vehicle performance, battery and charging, dimensions and capacity, and configuration sections. Numerical features use numeric controls, while categorical features use dropdowns.

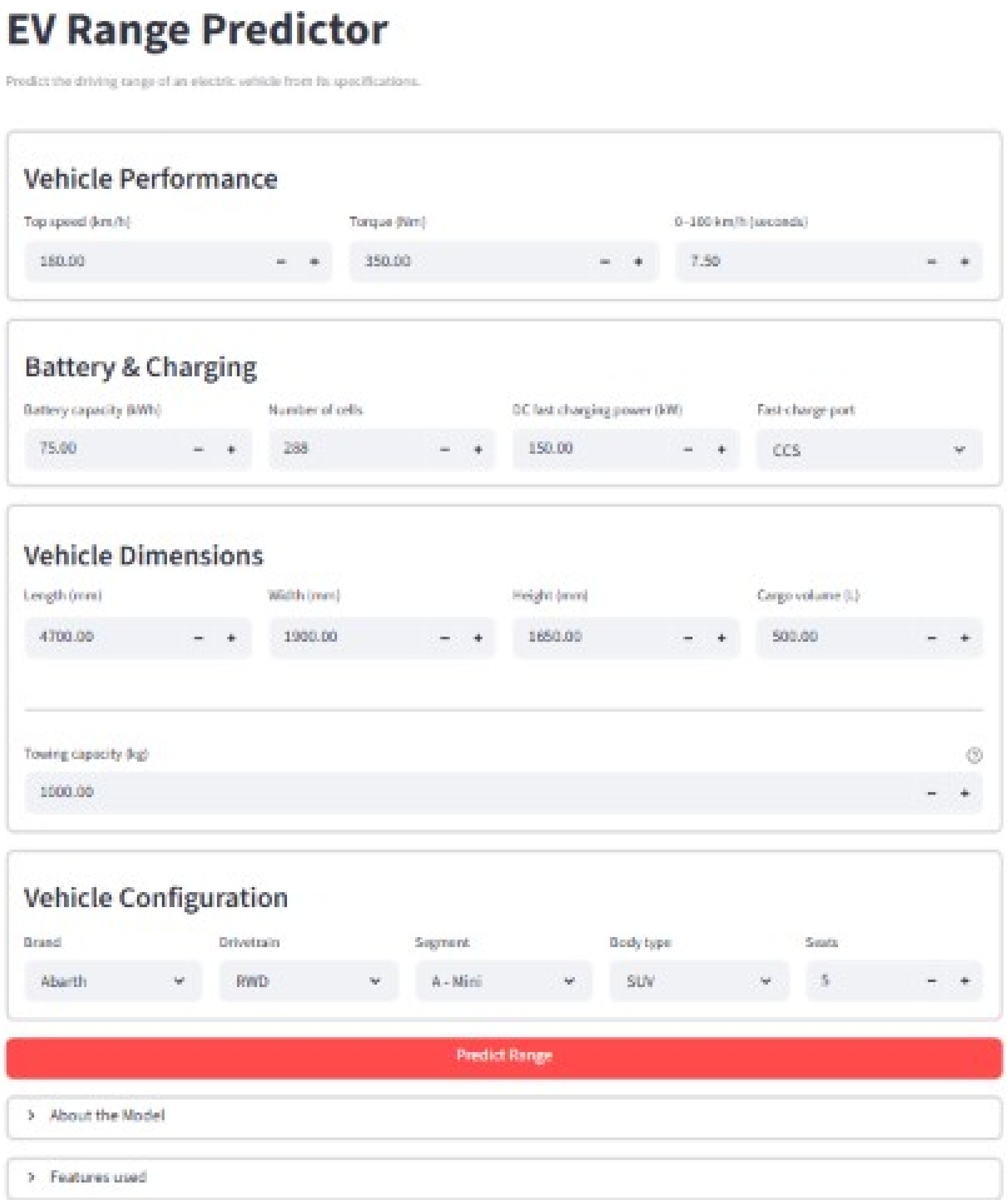

Figure 5. EV Range Predictor - Streamlit application interface.

The application loads `ev_range_svr_pipeline.joblib` directly. Because the saved artifact includes preprocessing and the trained SVR, the application does not manually re-encode or scale user inputs. The output is displayed as an estimated driving range in kilometres.

Interactive prediction flow

User enters 17 EV specifications -> pandas DataFrame -> saved preprocessing + SVR pipeline -> estimated range in km

14. Deployment Readiness

The project was prepared for Streamlit Community Cloud deployment. The local project uses relative model-file loading, and the dependency file pins the scikit-learn version required by the saved model artifact (scikit-learn 1.6.1).

Expected repository structure:

- app.py
- ev_range_svr_pipeline.joblib
- requirements.txt
- README.md
- .gitignore
- EV_Range_Prediction.ipynb
- Technical_Report.pdf

A representative local integration test using a Hyundai specification profile produced a prediction of approximately 465.76 km, consistent with the functioning saved pipeline and Streamlit interface. The deployed application should be treated as an estimate based on static specifications.

15. Limitations

- The model is trained on 478 EV records, which limits the amount of data available for learning highly complex patterns.
- Only static vehicle specifications are available; real-time driving conditions are not modeled.
- Weather, traffic, HVAC usage, road conditions, driving style, battery degradation, state of charge, and state of health are not included.
- Brand and segment effects may partly reflect correlated vehicle design and market characteristics rather than causal effects.
- The reported metrics describe the available dataset and held-out test split; real-world performance may differ on unseen vehicle designs or different operating conditions.

16. Future Scope

- Integrate real-world telemetry and energy-consumption data.
- Add weather, traffic, road-gradient, and HVAC-related variables.
- Include battery health and degradation indicators.
- Evaluate prediction intervals or uncertainty estimates in addition to point predictions.
- Expand the dataset with newer EV models and geographically diverse testing cycles.
- Monitor deployed performance and periodically retrain the model when sufficient new data becomes available.

17. Conclusion

A complete machine learning pipeline was developed for EV driving-range prediction from static vehicle specifications. The project combined systematic data cleaning, leakage prevention, exploratory analysis, feature selection, reproducible preprocessing, multiple regression models, hyperparameter tuning, final test evaluation, residual analysis, explainability, and an interactive Streamlit application.

The selected SVR model achieved an MAE of 10.309 km, RMSE of 13.646 km, and R^2 of 0.9824 on the held-out test set. Although Gradient Boosting achieved a slightly lower MAE, SVR provided better RMSE, R^2, and maximum-error performance. This makes SVR the preferred final model under a balanced objective that values both typical prediction accuracy and control of larger errors.

The final model artifact `ev_range_svr_pipeline.joblib` and Streamlit application provide a reproducible path from vehicle specifications to an estimated range prediction. The solution remains limited by its static-data scope and should be interpreted as an estimate rather than a guaranteed real-world driving range.

18. References

- Competition problem statement and dataset supplied for TECHTRACK 3.0 - EV-Focused ML Case Battle.
- Scikit-learn documentation for preprocessing, pipelines, model evaluation, SVR, and permutation importance.
- XGBoost documentation for regression and hyperparameter configuration.
- Streamlit documentation for application development and Community Cloud deployment.

End of Technical Report